

Problem 1

Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that

$$|x_n - x_m| < 1 \quad \text{for all } n, m \geq N_0.$$

Fix $m = N_0$. Then for all $n \geq N_0$,

$$|x_n| \leq |x_{N_0}| + 1.$$

Define

$$M_x = \max \{|x_1|, \dots, |x_{N_0}|, |x_{N_0}| + 1\}.$$

Then $|x_n| \leq M_x$ for all $n \in \mathbb{N}$. Thus (x_n) is bounded.

Similarly, since (y_n) is Cauchy, there exists $M_y > 0$ such that

$$|y_n| \leq M_y \quad \text{for all } n \in \mathbb{N}.$$

Let $\varepsilon > 0$ be given. For arbitrary $n, m \in \mathbb{N}$,

$$|z_n - z_m| = |x_n y_n - x_m y_m|.$$

Add and subtract $x_m y_n$:

$$|x_n y_n - x_m y_n + x_m y_n - x_m y_m|.$$

Factor:

$$|y_n(x_n - x_m) + x_m(y_n - y_m)|.$$

By the triangle inequality,

$$|z_n - z_m| \leq |y_n||x_n - x_m| + |x_m||y_n - y_m|.$$

Using boundedness,

$$|z_n - z_m| \leq M_y |x_n - x_m| + M_x |y_n - y_m|.$$

Since (x_n) is Cauchy, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x_m| < \frac{\varepsilon}{2M_y} \quad \text{for all } n, m \geq N_1.$$

Since (y_n) is Cauchy, there exists $N_2 \in \mathbb{N}$ such that

$$|y_n - y_m| < \frac{\varepsilon}{2M_x} \quad \text{for all } n, m \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then for all $n, m \geq N$,

$$M_y |x_n - x_m| < \frac{\varepsilon}{2} \quad \text{and} \quad M_x |y_n - y_m| < \frac{\varepsilon}{2}.$$

Therefore,

$$|z_n - z_m| \leq M_y |x_n - x_m| + M_x |y_n - y_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|z_n - z_m| < \varepsilon \quad \text{for all } n, m \geq N.$$

By definition, a sequence (x_n) in $\mathbb{R}$ is called *Cauchy* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon \quad \text{for all } n, m \geq N.$$

Since we have shown that (z_n) satisfies this condition, it follows that (z_n) is a Cauchy sequence. □

Problem 2

a)

Consider the sequence

$$(x_n) = \frac{1}{2}, \frac{1}{3}, -\frac{2}{3}, \frac{1}{4}, -\frac{2}{4}, \frac{3}{4}, \dots$$

Boundedness. Observe that each term satisfies

$$-1 \leq x_n \leq 1 \quad \text{for all } n \in \mathbb{N}.$$

Indeed, the numerator is always less than or equal to the denominator in absolute value, so $|x_n| \leq 1$. Therefore, the sequence is bounded both above and below, which means (x_n) is bounded.

Non-monotonicity. Consider the first few terms:

$$x_1 = \frac{1}{2}, \quad x_2 = \frac{1}{3}, \quad x_3 = -\frac{2}{3}, \quad x_4 = \frac{1}{4}.$$

- $x_2 < x_1$, so the sequence is not monotonically increasing.
- $x_4 > x_3$, so the sequence is not monotonically decreasing.

Hence, the sequence (x_n) is bounded but **not monotonous**.

b)

Recall that a sequence (x_n) is Cauchy if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } |x_m - x_n| < \varepsilon \quad \forall m, n \geq N.$$

Consider the sequence

$$x_n = \frac{1}{2}, \frac{1}{3}, -\frac{2}{3}, \frac{1}{4}, -\frac{2}{4}, \frac{3}{4}, \dots$$

Define two subsequences:

$$\begin{aligned} x_{n_k}^+ &= \text{largest positive term in the } k\text{-th block} = \frac{1}{2}, \frac{1}{4}, \frac{1}{5}, \dots \\ x_{n_k}^- &= \text{largest negative term in the } k\text{-th block} = -\frac{2}{3}, -\frac{1}{2}, \dots \end{aligned}$$

Observe that for all $k \in \mathbb{N}$,

$$|x_{n_k}^+ - x_{n_k}^-| \geq \frac{1}{2}.$$

Take $\varepsilon = \frac{1}{2}$. Then, for any $N \in \mathbb{N}$, we can choose $m = n_k^+ \geq N$ and $n = n_k^- \geq N$, so that

$$|x_m - x_n| = |x_{n_k}^+ - x_{n_k}^-| \geq \varepsilon.$$

This shows that no matter how large N is chosen, the terms do not all get within ε of each other. Therefore, (x_n) is not a Cauchy sequence. In addition, the sequence is bounded, but not monotonous, therefore it does not converge, and hence also not a Cauchy sequence. $\square$

c)

Consider the sequence

$$x_n = \frac{1}{2}, \frac{1}{3}, -\frac{2}{3}, \frac{1}{4}, -\frac{2}{4}, \frac{3}{4}, \dots$$

Strictly increasing subsequence. Define a subsequence (y_n) by taking the largest positive term from each block:

$$y_1 = x_1 = \frac{1}{2}, \quad y_2 = x_6 = \frac{3}{4}, \quad y_3 = x_{12} = \frac{5}{6}, \quad y_4 = x_{20} = \frac{7}{8}, \dots$$

Expanded, the first few terms are:

$$y_n = \frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \frac{7}{8}, \frac{9}{10}, \dots$$

Clearly, $y_1 < y_2 < y_3 < y_4 < \dots$, so (y_n) is strictly increasing.

Strictly decreasing subsequence. Define a subsequence (z_n) by taking the first positive term from each block:

$$z_1 = x_1 = \frac{1}{2}, \quad z_2 = x_2 = \frac{1}{3}, \quad z_3 = x_4 = \frac{1}{4}, \quad z_4 = x_7 = \frac{1}{5}, \dots$$

Expanded, the first few terms are:

$$z_n = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \dots$$

Clearly, $z_1 > z_2 > z_3 > z_4 > \dots$, so (z_n) is strictly decreasing.

d)

Strictly increasing subsequence Since (x_n) is bounded above by 1, we have

$$x_n \leq 1 \quad \text{for all } n \in \mathbb{N}.$$

Because (y_n) is a subsequence of (x_n) , it follows that

$$y_n \leq 1 \quad \text{for all } n \in \mathbb{N}.$$

Thus 1 is an upper bound for (y_n) , and we test 1 as a candidate limit.

For $n \geq 1$,

$$y_n = \frac{2n-1}{2n} = 1 - \frac{1}{2n}.$$

Let $\varepsilon > 0$ be given. Then

$$|y_n - 1| = \left| -\frac{1}{2n} \right| = \frac{1}{2n}.$$

We require

$$\frac{1}{2n} < \varepsilon,$$

which is equivalent to

$$n > \frac{1}{2\varepsilon}.$$

Choose

$$N := \left\lceil \frac{1}{2\varepsilon} \right\rceil.$$

Then for all $n \geq N$,

$$|y_n - 1| = \frac{1}{2n} \leq \frac{1}{2N} < \varepsilon.$$

Hence,

$$\lim_{n \rightarrow \infty} y_n = 1.$$

Strictly decreasing subsequence For all $n \geq 1$, we have

$$z_n = \frac{1}{n+1} > 0.$$

Thus (z_n) is bounded below by 0. Moreover, as n increases, the denominator $n+1$ increases, so the values $\frac{1}{n+1}$ become smaller and approach 0. This suggests testing $z = 0$ as a candidate limit.

Let $\varepsilon > 0$ be given. Then

$$|z_n - 0| = \left| \frac{1}{n+1} \right| = \frac{1}{n+1},$$

since $n+1 > 0$.

We require

$$\frac{1}{n+1} < \varepsilon,$$

which is equivalent to

$$n > \frac{1}{\varepsilon} - 1.$$

Choose

$$N := \left\lceil \frac{1}{\varepsilon} - 1 \right\rceil.$$

Then for all $n \geq N$,

$$|z_n - 0| = \frac{1}{n+1} \leq \frac{1}{N+1} < \varepsilon.$$

Thus,

$$\lim_{n \rightarrow \infty} z_n = 0.$$

Difference in limit of subsequences We have shown that

$$\lim_{n \rightarrow \infty} y_n = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} z_n = 0.$$

Since $1 \neq 0$, the two subsequences converge to different limits. However, if a sequence converges, then every subsequence must converge to the same limit. Therefore, (x_n) cannot converge.

$\therefore (x_n)$ diverges. $\square$

Problem 3

a)

Let $m, n \in \mathbb{N}$ with $m > n$. By the triangle inequality,

$$|x_m - x_n| = |(x_m - x_{m-1}) + (x_{m-1} - x_{m-2}) + \cdots + (x_{n+1} - x_n)| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k|.$$

Using the given bound $|x_{k+1} - x_k| < \frac{1}{k^{5/3}}$, we obtain

$$|x_m - x_n| < \sum_{k=n}^{m-1} \frac{1}{k^{5/3}}.$$

Since $\sum_{k=1}^{\infty} \frac{1}{k^{5/3}}$ is a convergent p -series with $p = 5/3 > 1$, its tails converge to 0. Therefore, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\sum_{k=n}^{\infty} \frac{1}{k^{5/3}} < \varepsilon \quad \text{for all } n \geq N.$$

Consequently, for all $m > n \geq N$,

$$|x_m - x_n| < \sum_{k=n}^{m-1} \frac{1}{k^{5/3}} \leq \sum_{k=n}^{\infty} \frac{1}{k^{5/3}} < \varepsilon.$$

Hence, (x_n) is a Cauchy sequence. Since every Cauchy sequence in $\mathbb{R}$ converges, it follows that (x_n) is convergent.

b)

Consider the sequence

$$s_n = \ln(n), \quad n \in \mathbb{N}.$$

Then

$$|s_{n+1} - s_n| = |\ln(n+1) - \ln(n)| = \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n} < \frac{1}{n^{3/5}} \quad \text{for all } n \geq 1,$$

where we used the standard inequality $\ln(1+x) < x$ for $x > 0$.

However,

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \ln(n) = \infty,$$

so the sequence (s_n) diverges. This example shows that even if the differences $|s_{n+1} - s_n|$ tend to grow slower, the sequence itself may fail to converge.

Problem 4

Suppose, for the sake of contradiction, that (x_n) does *not* converge to x . By the definition of convergence, this means that there exists some $\varepsilon_0 > 0$ such that for every $N \in \mathbb{N}$, there exists $n > N$ with

$$|x_n - x| \geq \varepsilon_0.$$

Equivalently, the set of indices

$$S := \{n \in \mathbb{N} : |x_n - x| \geq \varepsilon_0\}$$

is infinite.

Since S is infinite, we can enumerate its elements in increasing order:

$$S = \{n_1 < n_2 < n_3 < \dots\}.$$

Consider the subsequence (x_{n_k}) corresponding to these indices. Then, by construction,

$$|x_{n_k} - x| \geq \varepsilon_0 \quad \text{for all } k \in \mathbb{N}.$$

This subsequence cannot converge to x , because no term comes within ε_0 of x .

However, this contradicts the assumption that *every subsequence of (x_n) converges to x* .

Therefore, our assumption that (x_n) does not converge to x must be false. We conclude that

$$x_n \rightarrow x \quad \text{as } n \rightarrow \infty.$$

□

Problem 5

a)

For each $N \in \mathbb{N}$, define

$$X_N := \sup\{x_n : n \geq N\}, \quad Y_N := \sup\{y_n : n \geq N\}, \quad Z_N := \sup\{x_n y_n : n \geq N\}.$$

Then, by definition,

$$\limsup_{n \rightarrow \infty} x_n = \lim_{N \rightarrow \infty} X_N, \quad \limsup_{n \rightarrow \infty} y_n = \lim_{N \rightarrow \infty} Y_N, \quad \limsup_{n \rightarrow \infty} (x_n y_n) = \lim_{N \rightarrow \infty} Z_N.$$

Now, for all $n \geq N$, since $x_n \geq 0$ and $y_n \geq 0$, we have

$$0 \leq x_n \leq X_N, \quad 0 \leq y_n \leq Y_N \quad \implies \quad 0 \leq x_n y_n \leq X_N Y_N.$$

Taking the supremum over all $n \geq N$ gives

$$Z_N = \sup_{n \geq N} (x_n y_n) \leq X_N Y_N.$$

Finally, taking the limit as $N \rightarrow \infty$ and using the fact that the product of two convergent sequences is the limit of the product, we obtain

$$\limsup_{n \rightarrow \infty} (x_n y_n) = \lim_{N \rightarrow \infty} Z_N \leq \lim_{N \rightarrow \infty} (X_N Y_N) = \left(\limsup_{n \rightarrow \infty} x_n \right) \cdot \left(\limsup_{n \rightarrow \infty} y_n \right).$$

b)

Consider the sequences

$$x_n = (-1)^n - 10, \quad y_n = (-1)^n, \quad n \in \mathbb{N}.$$

Then

$$(x_n) = (-11, -9, -11, -9, -11, -9, \dots), \quad (y_n) = (-1, 1, -1, 1, -1, 1, \dots).$$

- The sequence x_n alternates between -11 and -9 , so

$$\limsup_{n \rightarrow \infty} x_n = -9.$$

- The sequence y_n alternates between -1 and 1 , so

$$\limsup_{n \rightarrow \infty} y_n = 1.$$

$$x_n y_n = ((-1)^n - 10) \cdot (-1)^n = (-1)^{2n} - 10(-1)^n = 1 - 10(-1)^n.$$

So the product sequence is

$$(x_n y_n) = (-9, 11, -9, 11, -9, 11, \dots).$$

The sequence alternates between -9 and 11 , so

$$\limsup_{n \rightarrow \infty} (x_n y_n) = 11.$$

$$(\limsup x_n) \cdot (\limsup y_n) = (-9) \cdot 1 = -9.$$

Clearly,

$$\limsup_{n \rightarrow \infty} (x_n y_n) = 11 > -9 = (\limsup x_n) \cdot (\limsup y_n).$$

Hence, the inequality from part (a) fails when negative terms are allowed.